Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/28/2006
PI Name:	Huso, David D.V.M., Ph.D.		
Project Title:	Chemoprevention of Radiation-Induced Neoplasms		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dhuso@jhmi.edu	**Fax:**	FY 443-287-2954
PI Organization Type:	UNIVERSITY	**Phone:**	443-287-2953
Organization Name:	The Johns Hopkins University School of Medicine		
PI Address 1:	600 North Wolfe Street		
PI Address 2:			
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21205	**Congressional District:**	7
Comments:	Deceased Jan 2016. (rje 11/16/16)		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	01/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-RE.002.05		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The major long-term risk associated with the types of low-dose radiation exposure that could possibly be encountered during prolonged space missions is radiation-induced cancers and they probably wouldn't appear until later in life. Proton and heavy ions as well as their secondaries are the major types of radiation likely to occur in the space environment. Chemopreventatives offer potential pharmaceutical countermeasures to reduce the increased cancer risks; however, no studies have tested chemopreventatives in studies relevant to proton or heavy ion-induced cancer. Surging scientific and public interest in applying chemoprevention strategies to people in the general population that have been identified to carry even slight increases in the genetic risk of developing cancer is fueling the identification of exciting new chemopreventive agents. Appropriate animal models provide a powerful tool for examining the effects of proton and heavy ion radiation at doses relevant to space as well as for directly evaluating the effectiveness of particularly promising chemopreventatives against cancers that occur following such exposures. Our project is a collaborative effort to characterize the carcinogenic risks associated with exposure to radiation which is relevant to the space environment		

	and to develop countermeasures to minimize the long term effects of radiation exposure. We are using a model of breast cancer, a tissue with known links between cancer development and radiation. Second we are characterizing countermeasures to this radiation-induced cancer using the prototype breast cancer chemopreventive tamoxifen to establish a proof of principle in the chemoprevention of cancer caused by protons and heavy ions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Chemoprevention of cancer in patients identified to be at increased risk for developing the disease is an emerging area of importance for the general population. As a result of the genomics revolution, the ability to detect genetic predisposition for even slightly increased risks of cancer is ever improving. Development of chemopreventives that mitigate this increased risk has not kept pace. This has resulted in a marked need for new effective and safe chemopreventive strategies for cancer. In particular, little is known about chemoprevention of cancers associated with radiation exposure due to occupational exposure, cancer treatments, or environmental exposure on earth. Our results are relevant to all of these important issues in the development of new chemopreventive strategies.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)